

Chiroptera: A Bio-Inspired Luxury Sailboat

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Understanding the Existir Problems

Aerodynamic and Performance Limitations

- Traditional sails operate efficiently only within a narrow wind range, losing lift under shifting or light winds.
- Fabric deformation and heeling forces reduce energy efficiency and directional stability.

Structural and Material Constraints

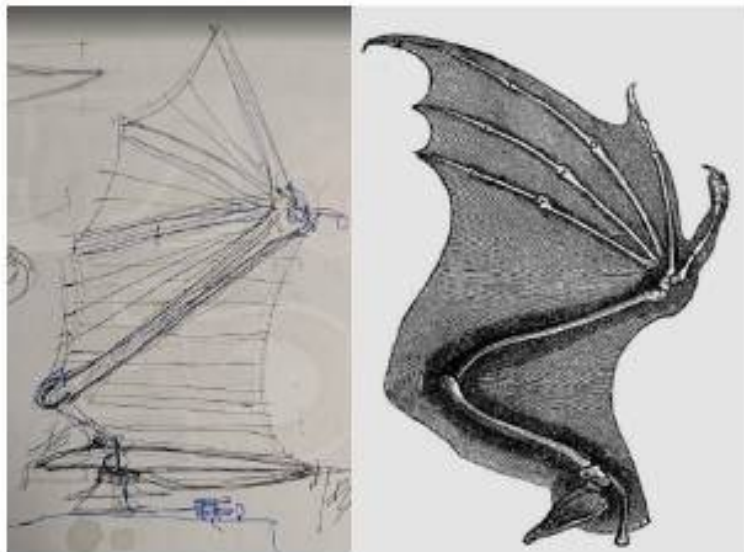
- Conventional rigs lack flexibility; rigid masts and soft sails cannot dynamically adjust to variable loads.
- UV, salt, and flex fatigue shorten lifespan; folding or reefing mechanisms add bulk and complexity.

Human, Control, and Integration Challenges

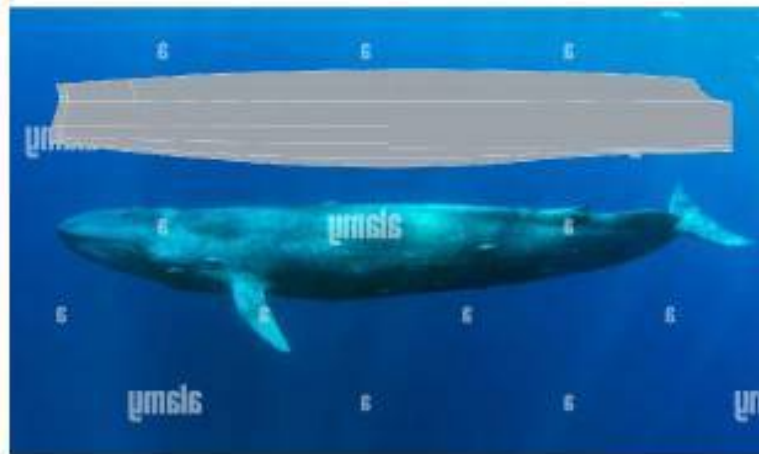
- Sail trimming remains manual and labor-intensive, requiring constant human input for performance.
- Poor aerodynamic-hydrodynamic coordination causes drag and instability during maneuvering.



Biological Inspiratio Summary



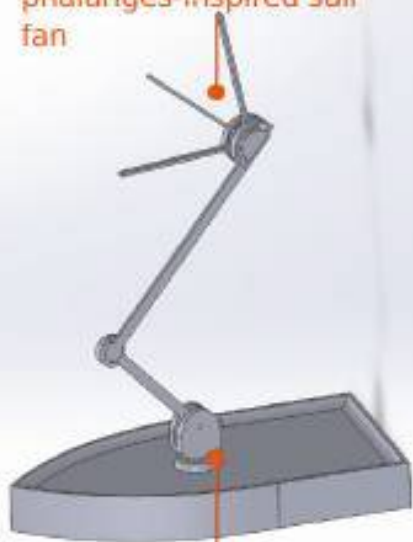
Bat wings combine lightweight skeletal ribs with tensioned membranes, allowing controlled deformation and high maneuverability. This anatomy informs the sail's rib-membrane system, enabling variable camber, smooth folding, and real-time aerodynamic adaptation.



Whales embody streamlined mass and gentle curvature that balance hydrodynamic efficiency with volumetric spaciousness. Their broad midsection and tapered ends inspired the hull's smooth flow profile and expanded interior volume for a stable, luxury-oriented form.

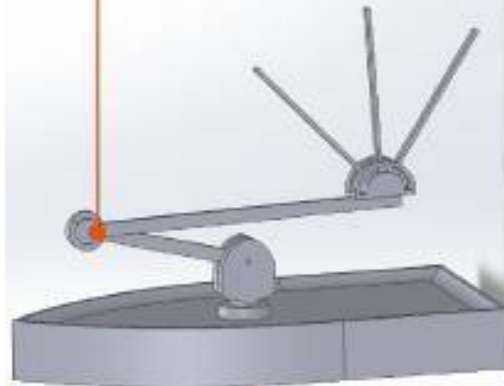
Initial Prototype

phalanges-inspired sail
fan

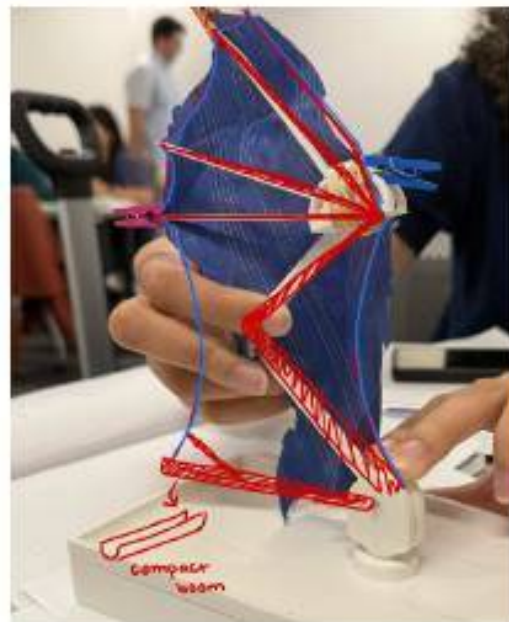


rotating
mast

collapsing
hinge



hull



Prototype hull & mast material: PLA Plastic
Prototype sail material: Cotton-Polyester Blend

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Theoretical Advantages of a Foldable Mast



Adaptive Aerodynamic Performance

Bio-inspired rib structures let the sail adjust curvature and stiffness in real time, optimizing lift and stability across varying wind speeds.

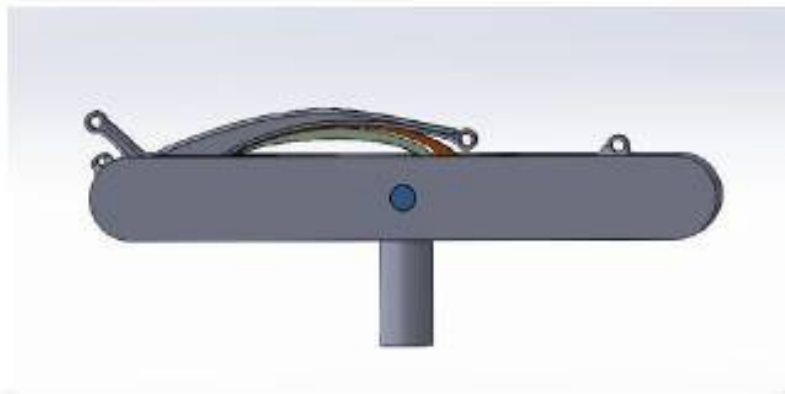
Storage & Logistics

Collapsed sail structure requires less space, Easier to transport the boat when folded

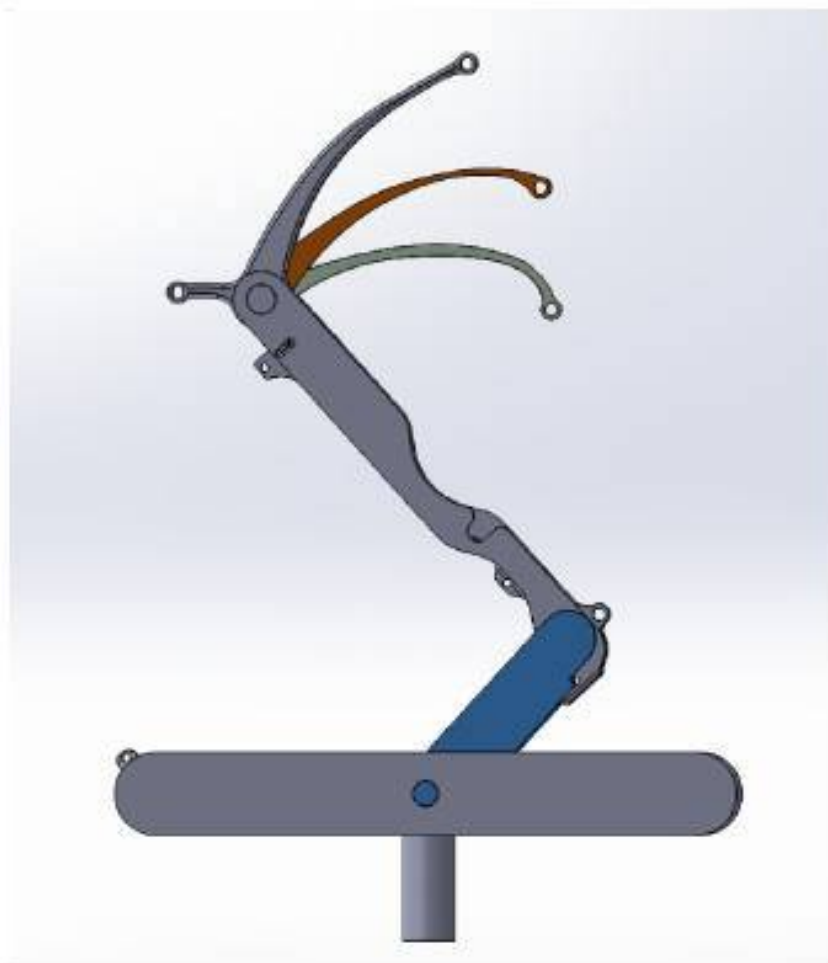
Rapid and Controlled Deployment

Origami-based folds enable fast, sequenced unfolding using minimal actuation, ensuring smooth, reliable sail deployment and quick adaptation to changing wind conditions.

Sail Design



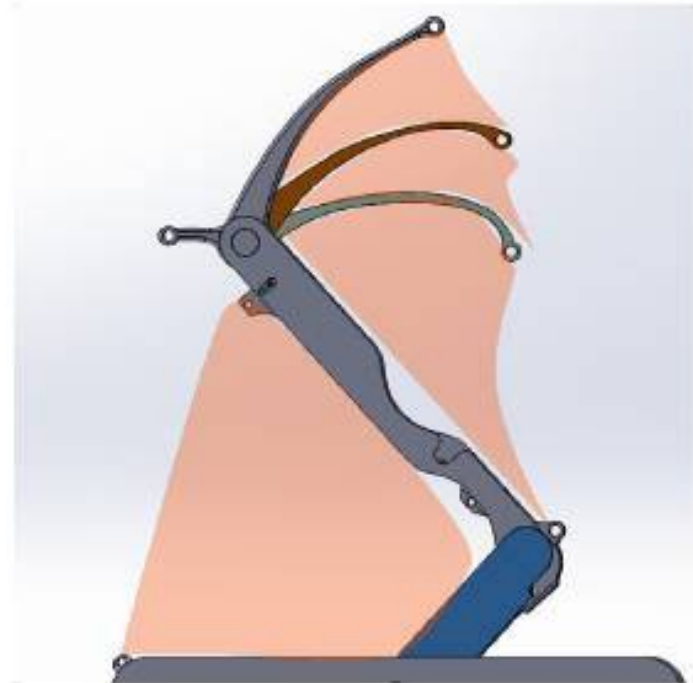
The mast folds on 3 joints, similar to the three joints of the bat wing. The mechanism will use a rope to extend the mast up and down via a pulley system connect to a winch. Improvements on this model are in development.



Sail Design: Skeletal Structure and Patagi Distribution



Bat-inspired ribs support flexible membranes, distributing stress evenly and allowing adaptive sail curvature for smoother aerodynamic performance. Wind tests will be done on the model to simulate its aerodynamics.



A diagram of a sperm whale (Physeter macrocephalus) in profile, facing left. The whale is dark grey with a lighter underbelly. A large, light-colored, elongated, and somewhat rectangular area is highlighted on the upper part of the whale's body, representing the location of the spermatophore. This area is outlined in red and has a light blue fill. The word 'spermatophore' is written in a stylized, light blue font across the middle of this highlighted area. The background is a solid light blue.

[illegible]

Hull Design: Current Progress and Roadmap

The current hull prototype requires further surface optimization to minimize hydrodynamic drag. **Minor surface irregularities will be refined in subsequent iterations to achieve smoother flow performance and improved efficiency.** Living area not included within current version

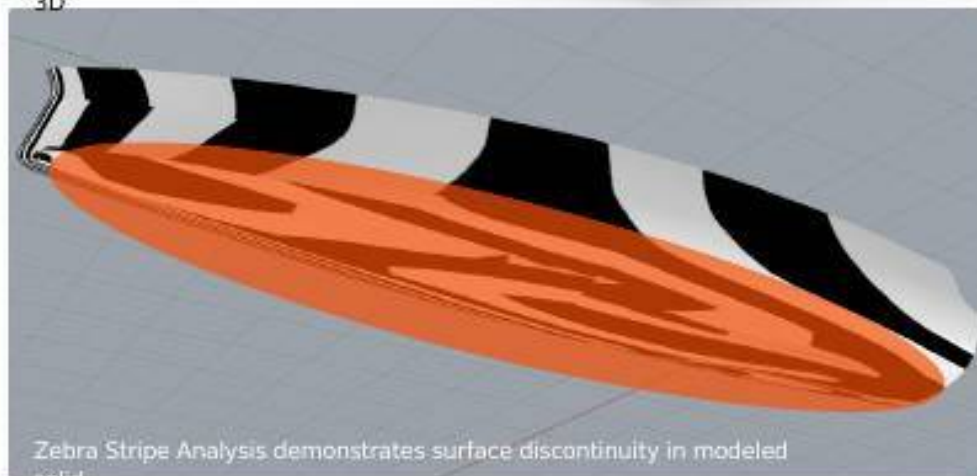


Minor surface quality issues are less visible via rendered view



Design Iterations of Hull Design Modeled in Rhino

3D



Zebra Stripe Analysis demonstrates surface discontinuity in modeled solid